

HW05 — Performance Testing Report

Student: NGUYỄN QUANG THÁI Student ID: 23127116 Class: 23KTPM2 Version / execution date: 1.0 / 2026-08-18

Executive summary

Apache JMeter 5.6.3 tested three independent EShop groups: read-heavy GET /api/products, auth-heavy POST /api/register, and transactional POST /api/admin/coupons. Load passed with p95 3 ms. Registration Stress had 0% errors but failed the working two-second latency gate at p95 2,473.40 ms. Coupon Spike p95 rose to 1,316.10 ms and returned to 17.80 ms. A ten-minute Products Endurance run sustained 481.001 post-ramp requests/s at p95 4 ms with no errors.

1. Scope and endpoint mapping

Scenario	Endpoint	Group	Rationale
Load	GET /api/products	Read-heavy	Repeated searches read product rows without state changes.
Stress	POST /api/register	Auth-heavy	Account-registration workflow creates identities.
Spike	POST /api/admin/coupons	Transactional	Authenticated coupon creation performs database writes.

Manual TODO: obtain the required endpoint/workflow uniqueness confirmation for the student group.

2. Environment and attribution

Item	Observed value
Host	LAPTOP-BKI58MTI, Lenovo 82JW
OS	Windows 11 Home Single Language 64-bit, build 26200.9168
CPU / RAM	Ryzen 5 5600H, 6 cores/12 logical; 16 GiB
Disk	Samsung NVMe SSD, about 476.94 GiB
Java / JMeter	Temurin 17.0.19+10 / JMeter 5.6.3
SUT	commit 85af3ba875c88283615e22cb108f13e2fccaf0e9 plus recorded dirty runtime DB

SUT and generator shared the laptop. Whole-system CPU/memory therefore cannot be attributed exclusively to Node/SQLite. No SUT source was changed.

3. Data and JMX design

Each group has a separate CSV and JMX plan. Registration emails receive a runtime UUID; coupon codes receive unique PERF_ values. Assertions verify expected HTTP behavior. Official tests ran non-GUI, wrote raw CSV JTL, HTML dashboard, JMeter log and resource CSV, and cleaned only generated records. Credentials were process-only and are absent from tracked plans/evidence.

4. Baseline and calibration

All three plan checks completed without assertion errors after startup effects were separated. Registration calibration showed 53.598 RPS at 25 users, 134.958 at 100, 135.065 at 200 and 137.503 at 400. p95 rose from 40.95 ms to 6,014.60 ms and 400 users produced 3.463% errors. This is an observed plateau near 135 RPS for that short profile—not a universal guarantee.

Products calibration produced 2,443.629, 2,518.079 and 2,529.031 RPS at 25/50/100 users, while p95 rose 14/26/50 ms. Fifty users was selected conservatively for Endurance.

5. Official results

Scenario	Samples	Error	Throughput	Average	p95	p99	Decision
Load	1,045	0.000%	8.914 RPS	1.61 ms	3 ms	3 ms	Pass
Stress	11,213	0.000%	183.877 RPS	1,515.37 ms	2,473.40 ms	2,802.76 ms	Fail latency
Spike burst	1,247	0.000%	140.302 RPS	390.91 ms	1,316.10 ms	1,540 ms	Pass
Endurance post-ramp	274,040	0.000%	481.001 RPS	2.11 ms	4 ms	9 ms	Pass for 10 min

Spike baseline/recovery p95 was 18.00/17.80 ms, demonstrating recovery in this run. Endurance final-window p95 rose from 2 to 8 ms and throughput fell 3.74% relative to the first window, but remained far within the latency objective with no failures.

6. Resource interpretation and threshold

Endurance backend working-set averages were 86.26 MB in the first and 86.46 MB in the last post-ramp minute. This does not show material backend-memory growth over ten minutes. Host-wide available memory and CPU were affected by JMeter/report generation. A provisional controlled-run gate is error rate <1%, p95 <2,000 ms, and Products post-ramp throughput ≥430 RPS; it must be re-baselined on a dedicated runner.

7. AI review and optimization

The preserved first pass was audited against raw labels/timestamps. It correctly separated errors from latency and Spike phases, but human review tightened workload comparability and causality. WAL and an email index are feasible Registration experiments. A coupon-code index is redundant because the column is UNIQUE; a generic pool does not fit a single SQLite handle; extra RAM and Redis lack current evidence. Parameterized product search is a security/correctness fix whose speed effect remains unproven.

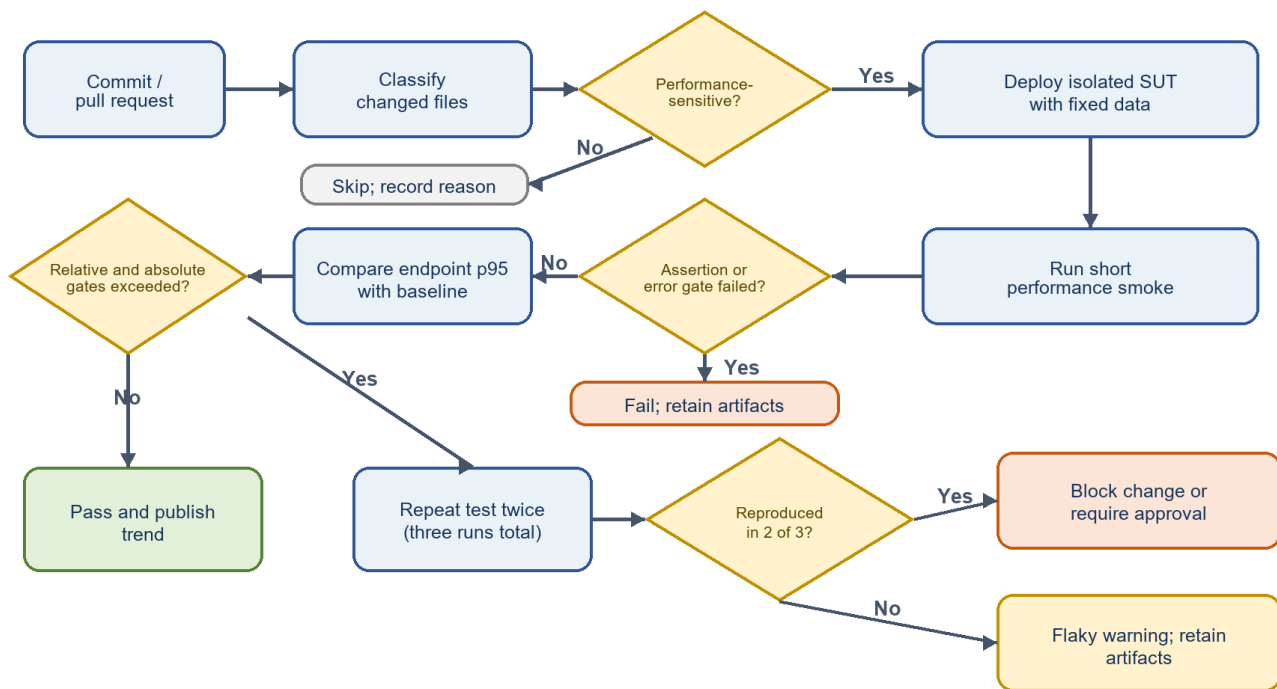
8. Reproduced issues

Dynamic smoke tests reproduced duplicate email acceptance and normal-user creation of Admin coupons. Registration saturation is documented as a performance issue without naming an unprofiled root cause. Markdown drafts exist under issues/; public Issue URLs remain manual TODOs.

9. Continuous testing and Agent Skill

The proposal defines changed-file filtering, an isolated fixed-data SUT, smoke/assertion gates, absolute and relative p95 gates, repeat-twice/two-of-three reproduction, pass/flaky/block paths and artifact retention. It is not implemented. The reusable Agent Skill passed the official validator and reproduced Spike burst and Endurance post-ramp values.

Continuous Performance Testing Flow



Proposed continuous performance testing decision flow

10. AI critique and disclosure

The 259-word critique explains why zero errors, aggregate phases and generic optimization advice require human checking. Full prompt-level records are in the AI Audit Report. Generative AI assisted execution, analysis and drafting; raw evidence, source and schema were used for verification.

11. Submission status

Completed: JMX/CSV, baseline/calibration, four official JTL/HTML/resource sets, analyses, proposal, critique, issue drafts, Agent Skill, environment and hardware screenshots.

Manual TODO: uniqueness evidence; simultaneous scenario screenshots; public repository URL; public Issue URLs; Vietnamese video ≥6 minutes plus Unlisted URL; final Audit signature. Until those are supplied, PDFs are drafts and no final ZIP should be created.

Hardware Evidence

DxDiag System screenshot captured on the test machine on 2026-08-18.

